

Department of Statistics and Computer Science
University of Peradeniya
CSC2021 – Programming using Data Structures
Labsheet 02
INTRODUCTION TO LINKED LISTS

- 1) Create a node with necessary functions and attributes for a linked list using an **integer** data type for the data component.
- 2) Create a linked list with the necessary attributes and methods.

The methods must address the following,

- a. Inserting a node to the **front**
- b. Inserting a node to the **rear**
- c. Inserting a node as the **next**
- d. **Searching** a node
- e. **Printing** the linked list

Note: Please consider all the scenarios when implementing the methods (Eg: Inserting a node to an empty list, handle empty list retrieval, etc)

- 3) Now perform the following operations in the driver class
 - a. Create a linked list with nodes having values 20, 30, 40 and 50 from head to tail in the specified order.
 - b. Insert node with value 10 to the front of the list
 - c. Insert a node with value 60 to the end of the list
 - d. Insert a node with value 70 next to the node with value 30
 - e. Display all the values in the list.
 - f. Delete node with value 70 in the list
 - g. Search for value 30 in the list and if it exists display “Value found” else display “Value Not Found”